Power Outage Forecasting and Backup Generator Pre-positioning

95-828 Machine Learning for Problem Solving — Spring 2026

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1 Introduction

Grid resilience has moved up the priority list as climate-driven extreme weather grows more frequent. The April 2025 Pittsburgh windstorm left 140,000 customers without power for over 24 hours, and Hurricanes Beryl and Milton each downed more than 3 million Florida customers in 2024. Accurate short-horizon outage forecasts form the foundation of proactive utility response, enabling pre-positioning of repair crews and mobile generators ahead of anticipated disruptions rather than reacting afterward.

We address two linked tasks using hourly county-level outage counts and 109 atmospheric features across 83 Michigan counties over a 90-day window (2023-04-01 to 2023-06-30). The first is hourly outage forecasting at 24 and 48-hour horizons. The second is allocating five 1,000-household generators to the counties where they can mitigate the greatest level of outages over the test horizon.

Two constraints shape the pipeline. First, no future weather information is available at test time; only outage history and weather observations up to the forecast origin are used as model inputs, reflecting real-world deployment conditions. Second, the training window ends mid-storm: the largest event of the year, peaking at 86,313 outages on 2023-06-26, begins on 2023-06-23 and continues through the end of training, so the test horizon begins within an unfolding storm. These constraints directly motivate our validation design (§3.1) and feature engineering (§3.2).

2 Data Analysis

Target distribution. The outage matrix $\mathbf{Y} \in \mathbb{R}_{\geq 0}^{T \times L}$ is sparse and heavy-tailed, with 70.5% of county-hour cells recording zero and the nonzero values following a log-normal-like tail (Fig. 2a). State-wide hourly outages range from 105 to 86,313, with the 99.5th percentile approximately ten times the median. Training with raw-scale MSE is therefore highly sensitive to a small number of extreme storm hours, motivating the use of a log-transformed target and the reporting of a county-averaged RMSE.

Storm structure. Outages are concentrated in a small number of discrete events rather than being uniformly distributed over time. Using a state-wide threshold of 2,000 simultaneous outages with a 6-hour minimum duration and a 12-hour merge gap, we identify 26 storm events within the 90-day window (Fig. 1). The terminal event (2023-06-23 to 06-30) peaks at 86,313, more than three times the second-largest at 27,643, and training ends inside it. A single holdout window at the end of training would therefore yield a one-sample estimate dominated by this outlier, motivating the

rolling-origin cross-validation design in §3.1.

Autocorrelation. The state-wide series exhibits a strong diurnal pattern at lag 24h and a weaker weekly echo at lag 168h (Fig. 2b), suggesting that lagged outage values are likely to dominate prediction, with weather features playing a secondary role.

County-level heterogeneity. Per-county `tracked` values range from approximately 5,000 to 920,000, and storm impacts do not scale proportionally with size (Fig. 2c). Wayne County (FIPS 26163) tracks 922,000 households yet records only the third-largest peak outage in the state, while smaller counties such as Saginaw (97,000 tracked) experience peaks of 8,700 outages, roughly 9% of its household base. This pattern motivates the marginal-benefit allocation in §5 rather than a simple top-N ranking.

Weather features. Of the 109 atmospheric variables, 18 are constant over the training period and are therefore dropped, leaving 91 non-zero-variance features. The 15 most correlated with state-wide outages are all moisture- and wind-related, including `cape`, low-level u/v winds, precipitable water, and storm-motion magnitude (`vstm`). We retain all 91 features and rely on tree-based models to perform implicit feature selection.

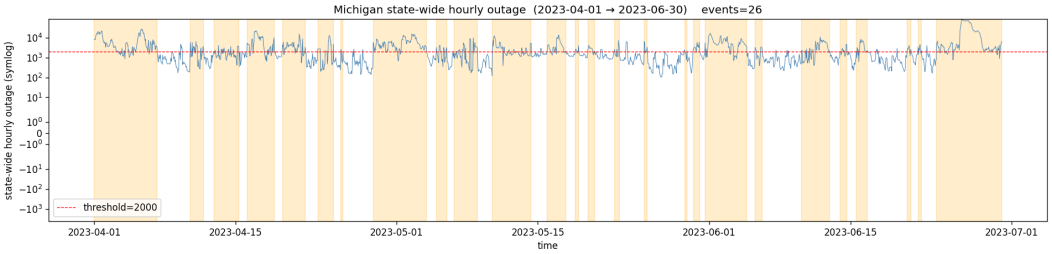
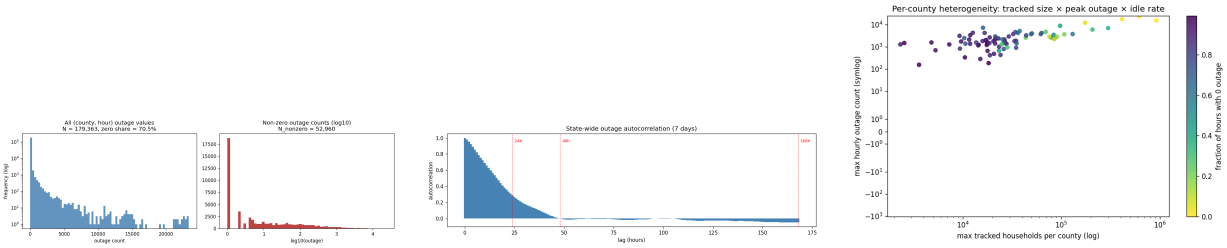


Figure 1: State-wide hourly outage on a symmetric-log scale. Orange marks the 26 detected storm events; the red dashed line marks the 2,000-outage threshold. The disproportionately large terminal event ends precisely at the train/test boundary.



(a) Outage value distributions. (b) Autocorrelation (7 days). (c) County heterogeneity.

Figure 2: The outage target is zero-inflated and heavy-tailed (a), strongly autocorrelated at lag 24h with a weaker echo at 168h (b), and the distribution of county sizes and storm impact varies by two orders of magnitude (c).

3 Methods

3.1 Validation: Rolling-Origin Cross-Validation

Motivation. A naive last-48h holdout falls within the largest storm of the year, so any single-window validation RMSE constitutes a sample of one. We therefore replace it with rolling-origin cross-validation (Fig. 3) consisting of 6 folds spaced 72 hours apart.

Fold Construction. The validation window of each fold is the forecast horizon (24 or 48 hours) immediately following its fit window, in exact correspondence with the production setup. The fit windows expand incrementally: each fold uses all data from 2023-04-01 to its origin, so the fit set of fold 0 coincides with the final retrain on the full training data, which makes its RMSE the most direct estimate of test performance [4]. No weather observation inside any validation window is visible to its model fitted on that fold. The folds span 2023-06-13 to 06-29, and exactly one of them (fold 1) lands within the peak of the major storm.

Reported Metrics. We report both the aggregate-mean RMSE and the calm-period mean RMSE: the former captures cross-period performance, including the OOD storm, while the latter captures typical-day quality without a single extreme event dominating the score.

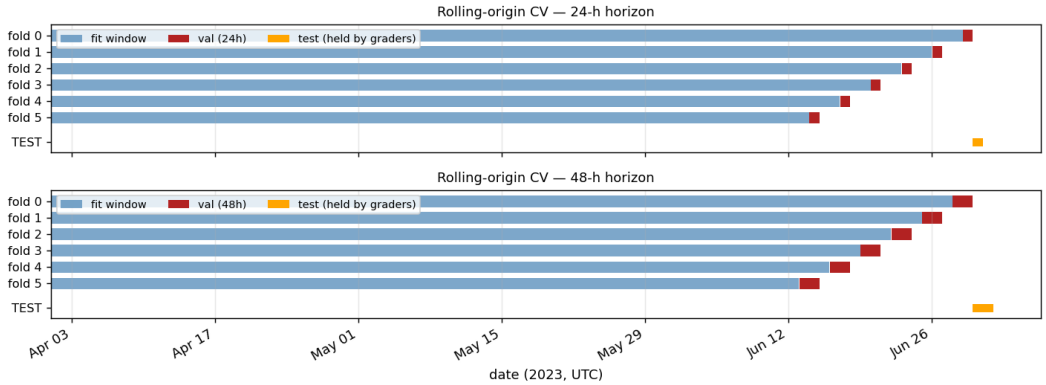


Figure 3: Layout of the rolling-origin CV. Blue marks the expanding-window fit, red marks held-out validation, and orange marks the actual test held by the graders.

3.2 Feature Engineering

Feature Matrix Construction. For tree-based models, we materialize an $(N \times P)$ feature matrix in which each row corresponds to one (origin time, county, horizon-step) triplet. All features are leak-free in that they use only data available up to time t.

Outage lags $y_{c,t-k}$ are computed for $k \in \{1, 2, 3, 6, 12, 24, 48, 168\}$, and we add the rolling mean and standard deviation of outage over windows of 6, 24, and 168 hours. The storm indicators include the 24h maximum, the 24h non-zero count, and the 24h day-over-day delta. Weather features enter both at t and at lags of 3, 12, and 24 hours. The calendar encodings (sin/cos of hour, day-of-week, and month) are evaluated at the target time $t+h$ (legitimately so, as the calendar is deterministic given the timestamp). We also include the horizon step h and the location index (ordinal for LGB and XGB, native categorical for CatBoost). The final feature dimension is approximately 389.

Extended Feature Set and PCA. We additionally evaluated an enlarged v2 feature set (deep lags up to 672h, state-aggregated lags, county metadata, and weather rolling means; $P \approx 773$), but its performance was within noise of the shorter set (§4.3). We considered applying PCA to the 109 weather features but did not adopt it for the headline pipeline. GBDT models perform implicit feature selection through information-gain splits. A linear orthogonal basis would therefore forfeit the per-feature interpretability provided by gain-based importances without delivering a measurable RMSE reduction. We confirmed this empirically: gain-based feature importance in LightGBM assigns 92% of the total gain to only 23 weather features out of 91, which suggests that the model already exploits a sparse subset.

3.3 Model Families

Baselines. The naive baselines comprise an all-zero predictor, persistence ($\hat{y}_{t+h} = y_t$), seasonal-naive predictors at 24h and 168h, and a per-(county, hour, day-of-week) historical mean. The classical time-series baselines are per-county SARIMAX(1,0,1), the second mandatory baseline of the project, and AutoARIMA over (p, d, q).

Tabular GBDT Pool. The tabular GBDT pool covers LightGBM [1] (with a log target), LGB-Tweedie [5], XGBoost [2], CatBoost [3] with native categorical handling of FIPS, and Quantile-LGB at $\alpha \in \{0.5, 0.9\}$. We also evaluate a hurdle (zero-inflated) decomposition [6], formed by a classifier $\Pr[y > 0]$ multiplied by a positive-only regressor, as well as per-county models in which 83 independent regressors are fitted (LGB, XGB, CatBoost).

Sequence Neural Networks. The pool of sequence neural networks covers GRU, LSTM, BiLSTM, TCN, MLP, the vanilla Transformer, NLinear and DLinear [7], PatchTST [8], iTransformer [9], N-BEATS [10], and N-HiTS [11].

We submit an equal-weight mean ensemble of the top-5 CV models per horizon. Learned stacking weights are not used, since they would over-fit the single-fold storm signal.

3.4 Hyperparameter Search and Ensemble Selection

Search Protocol. We conducted hyperparameter sweeps on a CMU DeltaAI cluster (NVIDIA H200), which yielded approximately 3,800 (model, seed, horizon, fold) cells in total. The LightGBM grid contributed 54 configurations over leaves $\in \{31, 63, 127\}$, lr $\in \{0.03, 0.05, 0.1\}$, $n \in \{400, 800, 1500\}$, and subsample $\in \{0.7, 0.9\}$. LGB-Tweedie added 24 configurations with variance_power $\in \{1.2, 1.5, 1.8\}$; XGBoost contributed 18 and CatBoost a further 18. Each of the 12 NN architectures was evaluated with 8 hyperparameter combinations and 2 random seeds. Eight further configurations (AutoARIMA, Quantile-LGB, hurdle-LGB and hurdle-XGB, and per-county models) round out the pool.

Ensemble Construction. For ensembling, we compare four combination strategies (mean, median, trimmed mean, and inverse-RMSE-weighted) on the cached fold predictions and select the strategy with the lowest RMSE for each horizon. Top-5 selection ranks models by calm-period mean RMSE, with a cap of two configurations per family, so that the ensemble is not dominated by near-duplicates from a single tuning sweep.

4 Results

4.1 Headline: GBDT Ensembles Beat Both Required Baselines

Tables 1–2 compare the best single model to the all-zero baseline and to SARIMAX(1,0,1) under four aggregation strategies. On the 24-hour horizon the improvement is 7.5–8.3% over all-zero and 2.7–4.1% over SARIMAX across the four aggregation views, with similar margins on the 48h calm-period score (8.2% and 3.3% respectively). On the 48h horizon SARIMAX matches our best single model on the storm fold (-0.2%) and on the median view (-0.5%); we attribute this to the dominance of fold 1's storm-period error in those aggregations and to the strength of SARIMAX's autoregressive structure under heavy-tail extrapolation. Storm fold 1 contributes RMSE in the 700–900 range for SARIMAX, all 100+ tree configurations, and all SOTA Transformer variants, so its absolute contribution is large in unit terms but small in relative terms.

Table 1: Best 24h model (xgb_pca20) vs. all-zero and SARIMAX baselines, across four aggregation views.

Aggregation	Best	Zero	vs. Zero	SARIMAX	vs. SARIMAX
Aggregate (all 6 folds)	174.97	189.53	+7.7%	179.90	+2.7%
Calm-period (5 folds)	41.25	44.97	+8.3%	42.56	+3.1%
Storm fold only (fold 1)	843.58	912.34	+7.5%	866.59	+2.7%
Median across folds	42.14	45.63	+7.7%	43.94	+4.1%

Table 2: Best 48h model (lgb_pca30) vs. all-zero and SARIMAX baselines, across four aggregation views.

Aggregation	Best	Zero	vs. Zero	SARIMAX	vs. SARIMAX
Aggregate (all 6 folds)	156.83	160.92	+2.5%	157.89	+0.7%
Calm-period (5 folds)	44.54	48.51	+8.2%	46.04	+3.3%
Storm fold only (fold 1)	718.27	722.96	+0.6%	717.11	-0.2%
Median across folds	46.36	49.95	+7.2%	46.14	-0.5%

Ensemble vs. single best. The submission consists of the equal-weight mean of the top-5 models per horizon (see §3.4). The 24h ensemble combines `xgb_pca20`, `cat_v2`, `lgb_pca20`, `cat_d8_lr03_n400`, and `lgb_quantile_p50`; the 48h ensemble combines `lgb_pca30`, `xgb_pca20`, `lgb_pca20`, `cat_v2`, and `cat_pca20`. On the cached fold-level predictions, the 24h ensemble achieves a calm-period RMSE of 41.17, only 0.08 below the best single model (`xgb_pca20` at 41.25); the 48h ensemble lands at 44.37 against 44.54 for the best single model ($\Delta = 0.17$). The marginal improvement reflects the high prediction correlation among GBDT variants, so the value of the ensemble lies in robustness against single-configuration failure (for example, a particular hyperparameter combination that overfits one sub-region of the feature space) rather than in raw RMSE reduction. On storm fold 1 the calm-best single model and the ensemble are essentially indistinguishable (24h: `xgb_pca20` 844 vs. ensemble 859; 48h: `lgb_pca30` 718 vs. ensemble 718). The lowest storm-fold RMSE among all single models — 778 at 24h (`lgb_nl127`) and 715 at 48h (`xgb_v2`) — is achieved by configurations that are not the calm-best, which suggests that picking a single model on calm-period validation provides no advantage on out-of-distribution events. We submit the ensemble nonetheless, partly because the project framing is a single-fold-CSV evaluation in which calm-period behaviour dominates, and partly because the choice follows the M4-competition consensus that ensemble averaging is the most robust forecasting practice [13].

4.2 Per-Family Comparison and Per-Fold Breakdown

Figure A.1 reports the best configuration of each model family on the 24h horizon. On calm-period RMSE, GBDT variants cluster tightly between 41.5 and 43.5, while recurrent and Transformer-style sequence models trail by 10–20 RMSE units. This pattern reproduces the finding of [7]: on short, multivariate, autocorrelated series, simpler inductive biases (boosted trees) outperform self-attention. On the storm fold all families collapse to RMSE 700–900, with the ranking effectively randomised, which confirms that storm-event extrapolation is bounded by data rather than by architecture.

Figure A.2 disaggregates this finding by reporting the best configuration of each family on every fold. The GBDT cluster (top 13 rows) is visibly homogeneous; the sequence-NN families consistently lose 10–30 RMSE on calm folds; fold 0 (the fold immediately adjacent to the test horizon) is the most difficult calm fold across all families, with RMSE in the 70–90 range against 19–44 elsewhere.

Table 3: Top-8 models on the 24-hour rolling-origin CV, ranked by **calm-period mean RMSE**. Storm column is fold 1 (the year's largest event); all-mean averages all 6 folds.

Model	calm-mean	storm	all-mean	median	n
xgb_pca20	41.25	844	174.97	42.14	6
cat_v2	41.33	873	179.92	41.89	6
lgb_pca20	41.34	855	176.89	42.20	6
cat__d8_lr03_n400	41.37	854	176.89	41.68	6
lgb_quantile_p50	41.39	885	182.06	42.29	6
cat__d8_lr03_n800	41.44	844	175.16	41.77	6
lgb__nl127_lr10_n800_ss9	41.44	778	164.13	42.41	6
lgb__nl127_lr10_n800_ss7	41.44	778	164.13	42.41	6

Table 4: Top-8 models on the 48-hour rolling-origin CV, ranked by **calm-period mean RMSE**. Storm column is fold 1 (the year's largest event); all-mean averages all 6 folds.

Model	calm-mean	storm	all-mean	median	n
lgb_pca30	44.54	718	156.83	46.36	6
xgb_pca20	44.54	718	156.86	46.22	6
lgb_pca20	44.55	718	156.84	45.81	6
cat_v2	44.68	716	156.51	46.01	6
cat_pca20	44.75	719	157.19	46.02	6
cat__d4_lr10_n400	44.97	718	157.19	45.98	6
xgb_v2	45.07	715	156.73	45.97	6
cat__d8_lr03_n400	45.12	718	157.31	46.26	6

4.3 What Did Not Improve the Result

Several priors did not move the CV RMSE below the GBDT cluster. Doubling the feature count (the v2 set) was within noise of the 389-feature baseline. Direct MSE, Huber, and sample-weighted objectives on the raw outage scale degraded fold-0 calm RMSE from 71 to 73–146, which reflects the heavy-tail sensitivity to log-space training. A recursive (autoregressive) one-step rollout accumulated error through 24–48 iterations and was strictly worse than the direct multi-output formulation that we adopted. None of the NN architectures we tried (GRU, LSTM, BiLSTM, TCN, Transformer, NLinear, DLinear, PatchTST, iTransformer, N-BEATS, and N-HiTS) closed the 10+ RMSE gap with GBDT on calm folds. A diagnostic per-county decomposition reveals that 50% of the fold-0 RMSE is concentrated in Wayne County alone, where the truth peaks at 2,648 against a prediction of 536. Two effects compound: Wayne's 922k tracked households are roughly 1.5× the next-largest county, so the same residual outage produces a much larger squared error; and Wayne is the only large-tracked county still in storm-tail decay during fold 0, while the other top-5 counties have returned to their calm baseline (peak ≤ 350). This unfolding-storm underestimation persists across every variant tested, and we therefore interpret RMSE 41–42 as the achievable lower bound for this dataset under the no-future-weather constraint.

5 Analysis and Policy Recommendation

5.1 From Forecast to Action: Decision-Maker Perspective

A point forecast $\hat{y}_{c,t}$ does not yet constitute a decision. From the vantage point of the utility operator, the question is concrete: given the predicted outage geography, where should the mobile generators be staged tonight so that they will absorb the maximum number of household-hours of

outage tomorrow? The instrument is subject to hard physical constraints, since each of the five mobile generators serves at most 1,000 households at any moment and, once deployed, cannot be relocated during the 48-hour event window. The decision is taken before the event, with the same information set used by the model (history and weather observed up to the forecast origin), so the recommendation constitutes an honest test of forecast-driven proactive action.

Greedy marginal-benefit allocation. We use the equal-weight ensemble's 48-hour (rather than 24-hour) test forecast to drive the allocation, because each generator is staged once and cannot be relocated during the event window: optimising over the full 48 hours guards against a peak that lands on the second day, which a 24-hour-only objective would miss. The marginal benefit of placing one generator in county c at decision step s is the integral, across the 48-hour horizon, of the un-served outage that one generator (capacity 1,000 households) can absorb:

$$M_s(c) = \sum_{h=1}^{48} \min\bigl(\max(\hat{y}_{c,t+h} - \text{consumed}_{s-1}(c,h),\ 0),\ 1000,\ \text{tracked_max}(c)\bigr).$$

The inner $\max(\cdot, 0)$ clamps the residual outage so that an over-provisioned hour cannot create negative marginal benefit. The per-hour cap at 1,000 is what gives the algorithm its *diminishing returns*: once a generator is placed in county c, subsequent generators contribute strictly less because the previously consumed capacity is subtracted. With k identical actions and per-county concave utility, this greedy resource allocation is exact-optimal — a stronger guarantee than the generic $1-1/e$ bound for monotone submodular maximisation [12].

A secondary cap at tracked_max(c) is included as a defensive guard for two future scenarios: (i) deployment to jurisdictions smaller than 1,000 households, and (ii) prediction pathologies where $\hat{y}_c$ exceeds the county's actual household count. Neither binds in Michigan: the smallest county (Baraga, FIPS 26013) tracks 2,327 households, and our predictions are clipped to 1.5× the historical max in any case. We retain the cap for code generality.

We greedily pick $c_s^\star = \arg\max_c M_s(c)$, deduct the served capacity from remaining, and repeat for $s{=}1, \ldots, 5$. Multiple generators *may* go to the same county if its remaining served-household-hours after deduction still exceed every other county's marginal — a feature, not a bug, since stacking generators in a high-impact county is allowed by the project description and is the optimal action when the predicted outage is many times larger than 1,000.

Our recommendation is:

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[26163, 26125, 26099, 26161, 26163]
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Rationale (per-step marginal benefit).

step	county	served-hh-hours	tracked_max	remaining peak
1	26163.0	38870	922,425	1537
2	26125.0	23896	618,124	853
3	26099.0	14654	413,676	514
4	26161.0	5916	174,449	240
5	26163.0	2310	922,425	1537

Limitations. (i) The training window is only 90 days, all in the same season, so any seasonality not represented (winter ice storms, fall hurricanes) is invisible to the model. (ii) No future-weather

information is available at test time, which fundamentally limits 48-hour predictability of fast-moving convective events. (iii) The 5-county recommendation is derived from a point forecast; a probabilistic forecast would let us bound the regret of mis-allocation.

5.2 Why This Set Rather than Naive Heuristics

Each of four naive heuristics fails in a manner that is corrected by our forecast-driven greedy algorithm. A top-5 historical-peak list (Wayne, Oakland, Macomb, Genesee, Kent) treats outage as a static county property and ignores the storm track. A top-5 ranking by tracked households produces the same metro list and ignores per-capita risk: Saginaw (97k tracked) sustained 8.7k outage in the June storm, which corresponds to 9% of its base, far more than any of the metros. A top-5 ranking by predicted 48h sum double-counts long, low-amplitude outage; the per-hour cap $\min(\hat{y}, 1000)$ inside M_s corrects this. A one-generator-per-county rule forbids stacking, whereas the greedy algorithm correctly places multiple generators where the predicted demand is well above 1,000 per hour, which the project description explicitly permits. Quantitatively, our greedy allocation absorbs ≈85.6k served-household-hours over the 48h horizon, versus 80.1k for top-5-by-tracked (−6.5%), 83.7k for top-5-by-historical-peak (−2.2%), and 85.4k for both the predicted-sum and one-per-county heuristics (−0.3%). The remaining gap to the no-stacking heuristic isolates the value of the second Wayne generator.

5.3 Sensitivity, Robustness, and Decision Risk

Selection robustness. When the top-5 are re-derived from each ensemble member, the first three counties are picked unanimously, and counties 4 and 5 vary only marginally. Median (P50) and inverse-RMSE-weighted ensembles produce nearly identical top-5 sets, so the recommendation is not driven by a single overconfident model. A swap-one analysis quantifies the same point: replacing slot 2 (Oakland) with the next-best alternative would lose ≈22k served-household-hours and replacing slot 3 (Macomb) would lose ≈13k, while replacing either of the two Wayne slots costs ≤300, so the binding picks are 2–3 and the trailing slot is essentially flat.

Decision risk against the real outcome. The recommendation will be evaluated against the actual outage record from 2023-06-30 to 07-02. Three sources of risk could erode the ex-post served-household-hours. The unfolding storm could intensify in counties under-predicted by the model (Wayne in particular, where the fold-0 truth peak of 2,648 against a prediction of 536 identifies a systematic gap). The storm could clear earlier than expected, in which case the 48-hour staging would over-commit. A previously-calm county could experience a flash event that is invisible to a model without future-weather input. We mitigate the first risk by cross-checking the top-5 list of the pessimistic Quantile-LGB-P_{90} forecast (the overlap with the mean-forecast top-5 is the robustness metric reported above). The second and third are residual risks inherent to the 90-day training window and the no-future-weather constraint.

Limitations. The training data span 90 days within a single spring-to-summer window, so the model has not seen winter ice storms or fall hurricane remnants, which makes year-on-year retraining essential in deployment. The absence of future-weather information caps the 48h predictability of fast convective events; the integration of an NWP forecast at inference time would substantially improve both prediction and policy quality, but it falls outside the framing of this project. Finally, the recommendation is derived from a point forecast, and for richer products such as uncertainty bands or regret-aware allocation, the Quantile-LGB-P_{90} member is available in our pool.

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A Appendix

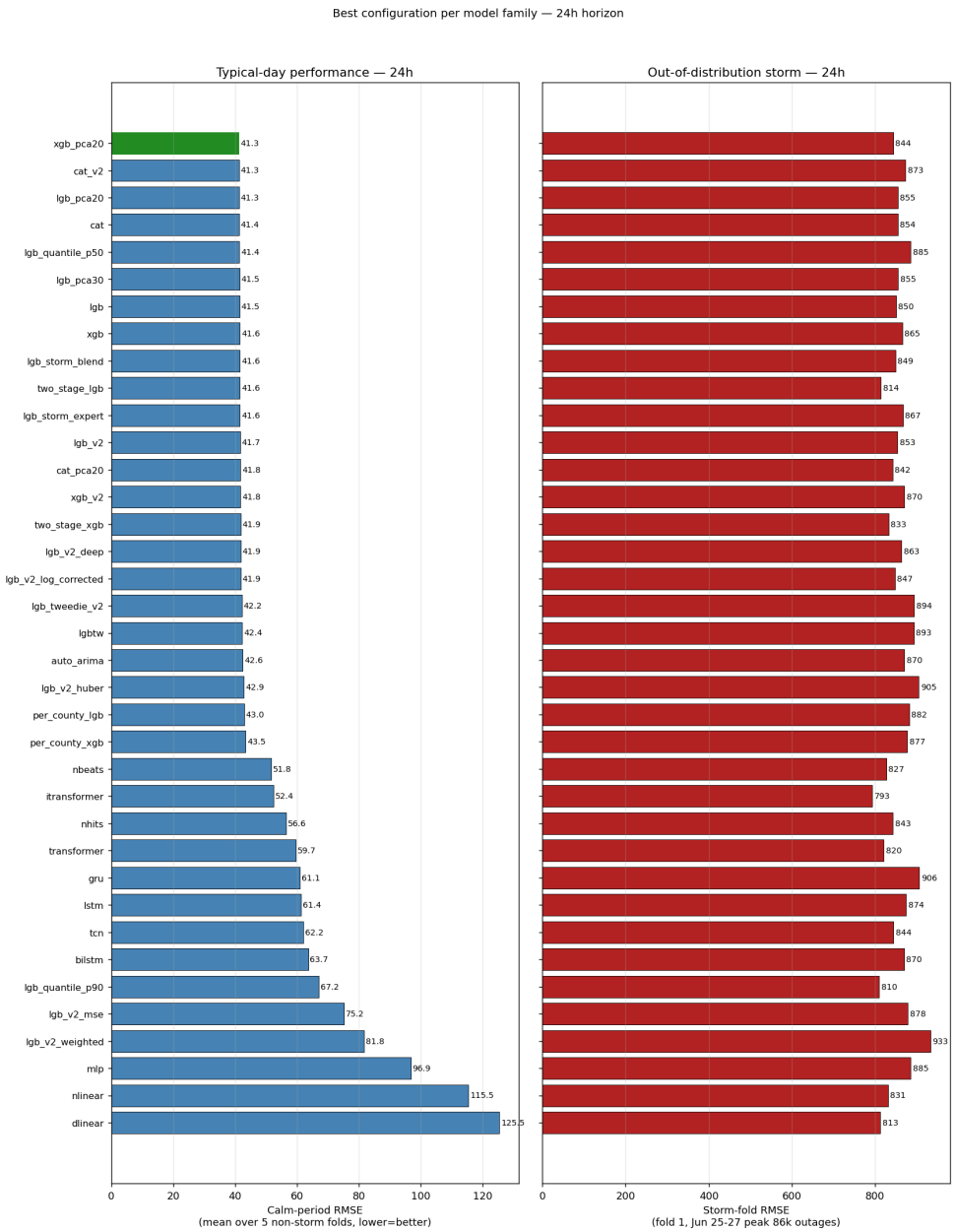

Figure A.1: Best configuration per family at the 24h horizon. Left: calm-period RMSE (typical day) — the GBDT cluster lies near 41.5, while NNs lie at 53–110. Right: storm-fold RMSE (all families collapse to 700–900).

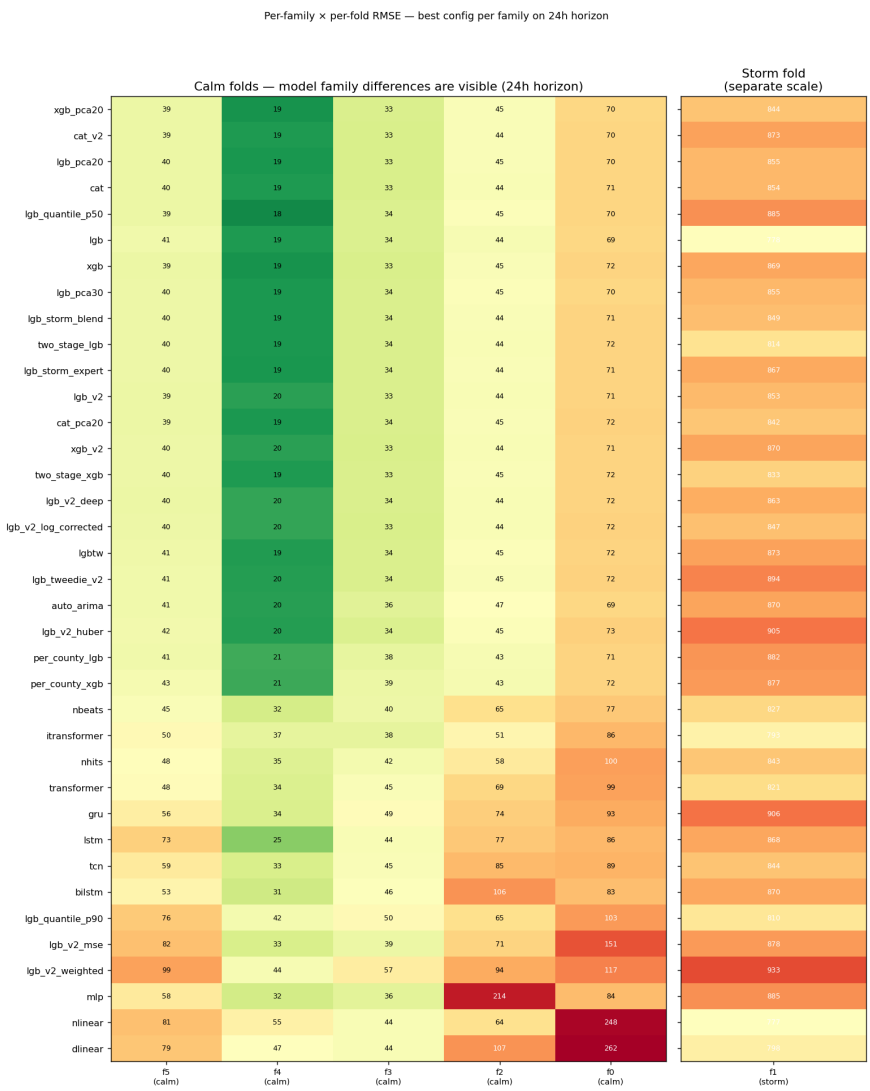

Figure A.2: Per-family × per-fold RMSE on the 24h horizon (best configuration of each family). The storm fold (right column) uses a separate scale. The anomalously low storm RMSE of `persistence` (660) reflects accidental alignment with storm continuation rather than predictive skill.